

TrES-3b

R-0520

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_240909 · created 2026-07-20T23:31:10+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460562.7135 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.055
Transit depth (Rp/Rs)^2	2.86 +/- 1.86 [%]
Orbital Inclination (inc)	77.78 +/- 2.67
Airmass coefficient 1 (a1)	1.075 +/- 0.02
Airmass coefficient 2 (a2)	-0.054 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	1.9 %
Best Comparison Star	None
Optimal Aperture	3.99
Optimal Annulus	9.4
Transit Duration (day)	0.013 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.169 $\pm$ 0.055 vs 0.16309 (deviation 0.1 σ)
Duration fit vs published	0.013 d vs 0.0567 d (deviation 1.93 σ)
Mid-time O–C	0.5 min
Depth SNR	3.0
Residual scatter	1.9 %

Light curve

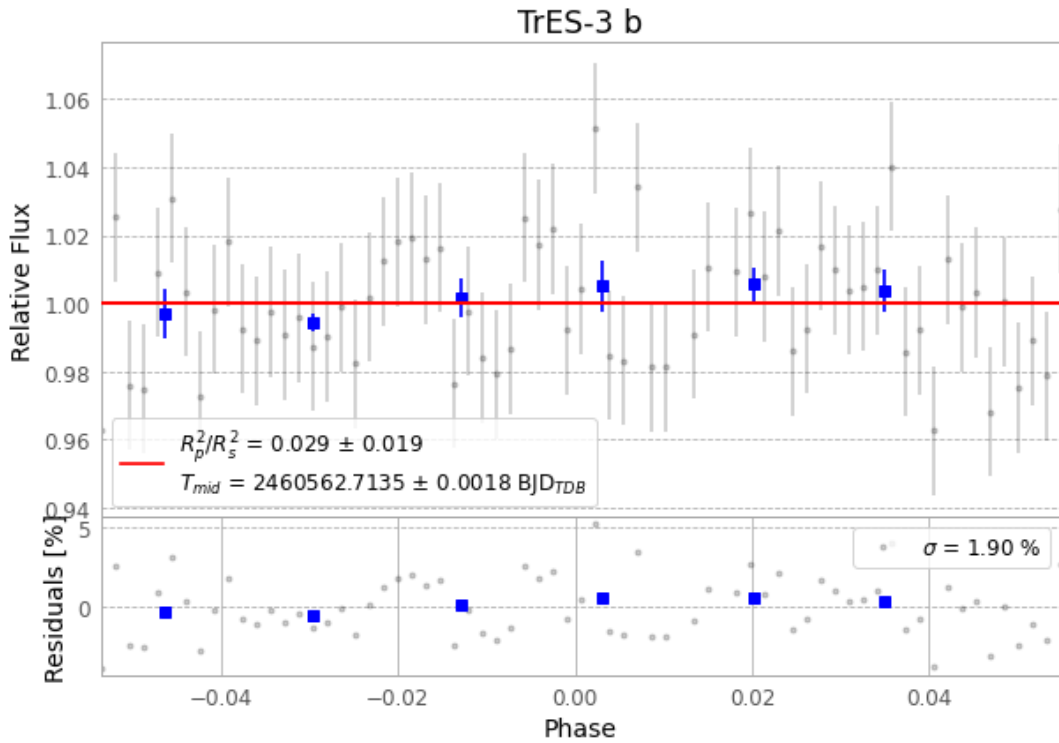


Figure 1 — FinalLightCurve_TrES-3 b_2024-09-08.png (SHA-256 717d7dce99809079...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [326, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a15ba78357a17b41...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

69 FITS frames (45 MB), TRES-3240909032416.FITS → TRES-3240909064815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-240909.log (b50139233c82...), FinalLightCurve_TrES-3 b_2024-09-08.png (717d7dce9980...), FinalLightCurve_TrES-3 b_2024-09-08.csv (caa1a38b4c23...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 23:31Z.